Spectral Features of PyroCb From Hyperspectral Infrared Sounders and Their Potential Use for PyroCb Remote Sensing

Xiaozhen Xiong, Xu Liu, Wan Wu, Liqiao Lei, Qiguang Yang, and Allen M. Larar

***Abstract*—Large wildfires and their induced thunderstorms, pyrocumulonimbus (pyroCb), are getting increasing interest and attention. Most previous studies are based on broadband satellite data, and the use of hyperspectral infrared (IR) data for pyroCb detection has not been explored. Australia's unprecedented fires at the end of 2019 to early 2020 induced an extreme pyroCb outbreak, and the largest event occurred on December 30, 2019 was successfully captured by the cross-track IR sounder (CrIS) onboard JPSS-1 and Suomi National Polar-orbiting Partnership (S-NPP). Analysis of these CrIS data demonstrated that the slopes and standard deviations (STDs) of the spectral brightness temperature (BT), derived from a linear fitting to the spectrum in 800–980 cm^{-1}, along with the split-window technique using selected CrIS channels, can aid in pyroCb detection. An inverted "V" feature in the spectra near 9.6 μm is associated with high pyroCb, and its depth, H_index, can provide information on cloud-top height. These characteristics are verified through radiative transfer model (RTM) simulations and analysis of another case of pyroCb induced from California Creek Fire on September 5, 2020. It is found that a combination of H_index and BT difference between 1231 and 677 cm^{-1} is promising to distinguish high pyroCb. A more accurate determination of pyroCb plume height can be achieved from the single field of view (SFOV) sounder atmospheric products (SiFSAPs), which indicate that several pyroCb pixels reach 3–6 km above the tropopause. These approaches could be applied to study more pyroCb using over 20 years of hyperspectral sounder data.**

***Index Terms*—Cloud-top height, fires, hyperspectral infrared (IR) measurements, pyrocumulonimbus (pyroCb), remote sensing, single-field-of-view retrieval.**

I. Introduction

WILDFIRES and their relation to weather, climate, and society are a topic of increasing interest and attention [1]. As a result of anthropogenic climate change, wildfires are becoming more frequent in many regions (e.g., [1], [2], [3], [4]). The wildfire-induced pollutants and smoke pose great risks to air quality and human health, and have a potential impact on climate (e.g., [5], [6], [7]).

Pyrocumulonimbus (pyroCb) refers to a peculiarly violent atmospheric disturbance linked to extremely intense wildland fire and manifested as a thunderstorm-like cloud (e.g., [1], [4]). This fire-induced and smoke-infused thunderstorm is among the most extreme and hazardous phenomena associated with intense wildfire episodes, and was found more frequently than previously recognized, with global annual pyroCb activity exceeding 40–50 distinct events and varying by a factor of two [4]. Many regions worldwide, particularly in the middle to high latitude regions, have experienced extremely active fire seasons and prolific pyroCb activity, including Canada, Alaska, eastern Siberia, and western U.S. (e.g., [8]). As the most vigorous form of biomass burning, pyroCb provide an efficient chimney-like smoke pathway for the transport of biomass-burning aerosols, gases, and cloud ice particles to reach the upper troposphere/lower stratosphere (UTLS) (e.g., [9], [10], [11], [12], [13], [14]). It is found that the magnitude of smoke plumes observed in the UTLS has increased significantly in recent years, rivaling or exceeding the impact from all volcanic eruptions observed over the last decade (e.g., [15], [16], [17]). Once in the stratosphere, the aerosol lifetime can extend from months to years and therefore can have a significant impact on the climate.

Australia's record-breaking temperature and extremely low precipitation caused unprecedented fire disasters at the end of 2019 to early 2020, resulting in significant damage, loss of properties, and human lives. The most intense fire and pyroCb activity were observed during Australian New Year Super Outbreak (ANYSO) in southeastern Australia; and 13 blowup fires were found to have contributions to the 38 distinct pyroCb pulses that can be divided into 18 distinct pyroCb [17]. The most extreme pyroCb activity occurred on December 30, 2019, which is also the first known occurrence of continuous pyroCb activity over an entire 24-h period and with some of the largest occurring at night or during the local morning hours [17]. The smoke plumes from Australia fires reached extraordinary altitudes of up to 35 km [13]. Enhancement of aerosol extinction was found in the southern hemisphere lower stratosphere; warm temperature anomalies of 1–2 K and ozone anomalies were also found after 2020 Australian bushfires.

The exceptional magnitude of ANYSO has initiated a new phase of pyroCb research aimed at understanding larger scales of potential atmospheric impacts and climate feedbacks (e.g., [18], [19], [20], [21], [22], [23], [24]). However, research

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on pyroCb is still a relatively new field, as pyroCb have rarely been observed before 2000 [1], [24]. By using modern satellite observations, it is found that some previous phenomena impacting stratospheric aerosols are likely relevant to some historic large wildfires and fire-induced pyroCb (e.g., [1]). Further research is required to quantify the future impact of pyroCb on personal safety, community threats, and global change, and to improve forecasting using observational and modeling approaches from the fire-scape level to global scale [4].

Polar-orbiting and geostationary sensors with multispectral capability have been employed to detect pyroCb (e.g., [10], [14], [25]). As the initial transport of the smoke cloud from fires, or young pyroCb plumes, is a peculiar mixture of smoke aerosols and water-ice, the particle sizes can be 1/3 to 1/2 of the normal cirrus crystals [26]. The lack of significant precipitation results in only a small fraction of the smoke being scavenged, and most of the smoke is exhausted through the anvil to the UTLS. Due to these unique microphysical signatures of pyroCb that differ from traditional convection, the difference of brightness temperature (BT) between a mid-IR channel near 3.7 μm (BT3) and BT11 has been used for the separation of pyroCb from other deep convection during daytime. For example, it was used in an automatic algorithm to identify deep convection near active fires using Geostationary Operational Environmental Satellites (GOES)-15 [25]; pyroCb have been observed by satellite occurring in the morning, afternoon, and in the middle of the night hours [1]. However, observation of pyroCb during nighttime is more challenging than at daytime as the information from these solar-impacted channels, such as BT3, cannot be used. Even though the split-window technique based on the BT difference (BTD = BT11–BT12) can be used to help detect pyroCb at nighttime (e.g., [26]), it is not an efficient way, and for example, the algorithm developed in [25] is not suitable for pyroCb detection if the entire life cycle of a pyroCb occurred after sunset.

Lecours et al. [27] first showed an infrared (IR) spectral atlas of polar mesospheric clouds, polar stratospheric clouds, cirrus clouds, stratospheric smoke from fires, and sulfate aerosols based on measurements from the Atmospheric Chemistry Experiment (ACE) satellite. Data from the ACE Fourier transform spectrometer (ACE-FTS) can provide altitude-resolved volume mixing ratio measurements for numerous atmospheric constituents, including many biomass-burning products [28], [29]. Using these retrieval products from ACE-FTS products, Boone et al. [28] and Zhang et al. [30] have investigated the impact on stratospheric compositions from the intense wildfires and pyroCb. However, to our knowledge, the hyperspectral IR spectral signatures for pyroCb have not been well documented before. As presented later this article, we found that during the largest pyroCb outbreak occurred on the evening of December 30, 2019, there were two overpasses captured by the cross-track IR sounder (CrIS) onboard the Suomi National Polar-orbiting Partnership (S-NPP) and Joint Polar Satellite System-1 (JPSS-1), or NOAA-20 satellites ~50 min apart [31], as well as one overpass of Atmospheric IR Sounder (AIRS) in between [32]. Using these unique observations from CrIS, we will first show some IR spectra of this pyroCb. Since the use of hyperspectral IR spectral information in pyroCb remote sensing has not been previously published, we will explore the potential to use these hyperspectral features for pyroCb detection. We will also use the retrieved single field of view (SFOV) sounder atmospheric products (SiFSAPs) [33] to analyze the pyroCb cloud-top height. A brief introduction of CrIS and other data used will be given in Section II. Some characteristics of pyroCb spectra from CrIS observations and radiative transfer model (RTM) simulations, as well as their potential use for pyroCb detection, are presented in Section III. Some discussion of the retrieved cloud properties and plume heights is also given in Section III. Section IV gives a summary and some conclusions.

II. Data and Method

CrIS is an advanced Fourier transform spectrometer that measures the thermal IR radiances in three spectral bands, i.e., the long-wave IR band 1 (648.75–1096.25 cm^{-1}), the mid-wave IR band 2 (1208.75–1751.25 cm^{-1}), and the short-wave IR band 3 (2153.75–2551.25 cm^{-1}) [31], [32], [33], [34], [35]. Since December 14, 2014, CrIS on S-NPP was switched to full spectral mode, which has a spectral resolution of 0.625 cm^{-1} in all three bands with 2211 channels in total. CrIS on J-1 (NOAA-20) was launched on November 18, 2017, and operated at full spectral mode. The nadir footprint of CrIS is ~15 km, and each CrIS on S-NPP or J-1 provides near-global coverage twice daily. To remove the ringing effects, as is done for the assimilation of CrIS data into Numerical Weather Prediction (NWP) models, the Hamming apodization is applied to the CrIS data used in this article [35], [36]. As the first space-borne nadir cross-track scanning IR spectrometer, AIRS was launched into a 705-km altitude polar orbit on the Earth Observing System (EOS) Aqua spacecraft on May 4, 2002, with 2378 channels covering from 649 to 1136, 1217–1613, and 2169–2674 cm^{-1} at high spectral resolution ($\lambda/\Delta\lambda$ = 1200) [32], [37]. The over 20 years of data from AIRS could be a valuable hyperspectral dataset for pyroCb study. However, due to performance degradation in certain channels over time [38], this article will only present the results using hyperspectral IR data from two CrIS instruments on S-NPP and NOAA-20.

To better map the location of pyroCb, the Visible IR Imaging Radiometer Suite (VIIRS) data collocated with CrIS will be used. VIIRS is an instrument onboard S-NPP and JPSS-1 that observes and collects global satellite observations that span the visible and IR wavelengths with 22 channels ranging from 0.41 to 12.01 μm [39], [40]. Five of these channels are high-resolution image bands or I-bands (with a spatial resolution of 375 m), and 16 serve as moderate resolution bands or M-bands (with a resolution of 750 m). The much higher spatial resolution of VIIRS than CrIS enables us to see some fine spatial structure of pyroCb.

To better use the information from so many hyperspectral channels, our first approach is to make a linear fitting to the BT in the spectral region from 760 to 980 cm^{-1} (or 13.2–10.2 μm) and then use the fit slopes and deviations for pyroCb detection; the latter is computed as the standard deviation (STD) between

the fit line and the BT from observations, and it becomes larger when water vapor line absorption is significant for some cases, such as clear sky or low water clouds. A previous study using AIRS data measured over the tropical ocean, Aumann et al. [41] discovered that, for some extremely cold cloud tops with BT1231 (8.1 μm) below 210 K, the observed spectra were inverted compared to clear sky cases, and this inversion could be identified using either strong CO_2 or water lines. In this article, we will explore a different approach by using the CrIS spectrum near the 9.6-μm O_3 band to identify spectral inversion associated with deep convection within pyroCb. Some deep convection cloud detection methods from [41] are examined for the potential pyroCb detection on the basis of some hyperspectral features derived from VIIRS-matched pyroCb pixels. The theoretical basis of these detection approaches can be further illustrated using radiative transfer simulations. Here, using a fast principal component (PC) radiative transfer model (PCRTM), we simulate the CrIS spectra for water and ice clouds with different cloud tops and effective diameters. The input atmospheric temperature, water vapor, and ozone profiles are taken from the fifth generation of European Center for ECMWF atmospheric reanalysis of the global climate (ERA5) [42]. ERA5 has a resolution of $0.25^\circ \times 0.25^\circ$ latitude-by-longitude grid with a 1-h interval. To match up with CrIS measurements, one profile near the center of the pyroCb is selected.

Finally, we will use the SiFSAP, developed by NASA using CrIS combined with Advanced Technology Microwave Sounder (ATMS) [33], to estimate the pyroCb top altitude above tropopause for each CrIS pixel with a spatial resolution of ~15 km at nadir. Traditionally, a few algorithms have been developed to generate atmospheric temperature, water vapor, O_3, and other trace gas products with a coarser spatial resolution than SiFSAP. These algorithms include the NOAA Unique Combined Atmospheric Processing System (NUCAPS) [43] and the Community Long-Term IR Microwave Combined Atmospheric Product System (CLIMPCAPS) [44], [45]. They both have some heritage from the original AIRS algorithm (e.g., [46], [47], [48]) that relies on the cloud clearing (CC) method to convert nine field of view (FOV) radiance spectra within one field-of-regard (FOR) into a cloud-cleared radiance spectrum, resulting in a spatial resolution of approximately 45 km at nadir. Different from the above two approaches, SiFSAP products are derived in each FOV without using the CC method, and the SFOV retrieval algorithm is based on PCRTM and an optimal estimation retrieval method [49], [50], [51], [52], [53], [54], [55]. As PCRTM includes accurate computation of cloud multiple scattering, which uses cloud lookup tables (LUTs) calculated by the 32-stream DIScrete Ordinates Radiative Transfer (DISORT) model [56], this algorithm retrieves cloud properties (phase, optical depth, droplet size, and cloud height) together with atmospheric temperature and water vapor profiles simultaneously. Another difference is that NUCAPS and CLIMCAPS use only a portion of the channels selected from the 2211 CrIS channels, whereas the SiFSAP algorithm uses information from all spectral channels by compressing all channel radiances into the PC domain. In general, the information content of a CrIS spectrum is captured by the first 120–160 PCs, and the retrievals are made in PC domain space using these PCs, which helps to reduce the apparent noise of the retrieval products and speed up the inversion process. The combined use of microwave sensor, i.e., ATMS, with CrIS allows for accurate retrievals under cloudy conditions. More details about SiFSAP can be obtained from Wu et al. [33]. Some advantages of using SiFSAP products for process-oriented analysis of the weather events, such as stratospheric intrusions and cold air outbreaks, have been previously demonstrated by Xiong et al. [57], [58]. The thermal tropopause, which is defined as the lowest level at which the (negative) vertical gradient in temperature decreases to 2 °C per km or less [59], is computed by adopting the method from [60]. The SiFSAP-retrieved temperature profiles for different pixels are used to compute their thermal tropopause heights, and its difference from the SiFAP-retrieved cloud-top height for each individual pyroCb pixel represents the pyroCb top altitude above tropopause.

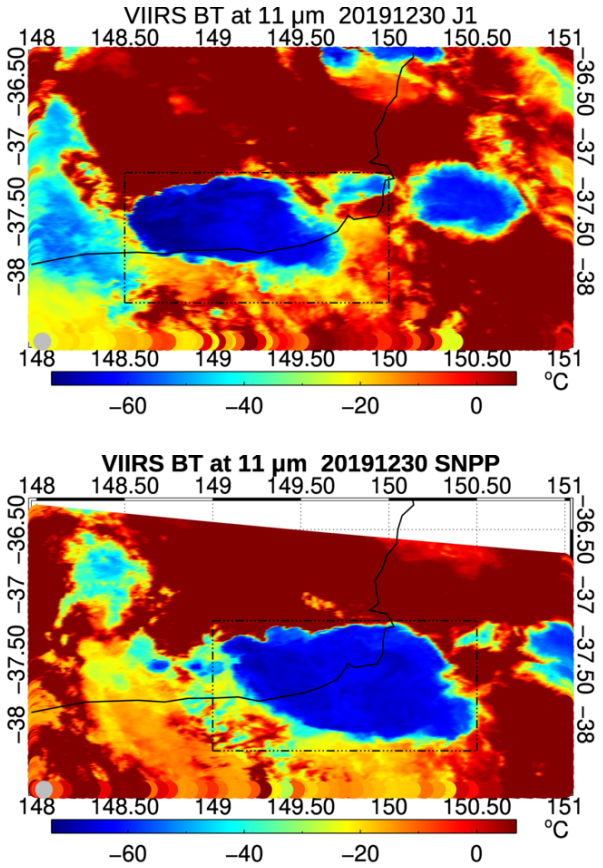


Fig. 1. BT at 11 μm for (top) VIIRS on J-1 (14:50 UTC) and (bottom) S-NPP (15:12 UTC) on December 30, 2019. The pyroCb location is marked by the squares with black dashed lines, indicating its movement of about 0.5° eastward and expansion toward southeast.

III. Results and Discussion

A. Characteristics of PyroCb Spectra

As indicated in previous studies, pyroCb can inject massive amounts of smoke into the UTLS. These smoke particles can play an important role in creating the extremely small

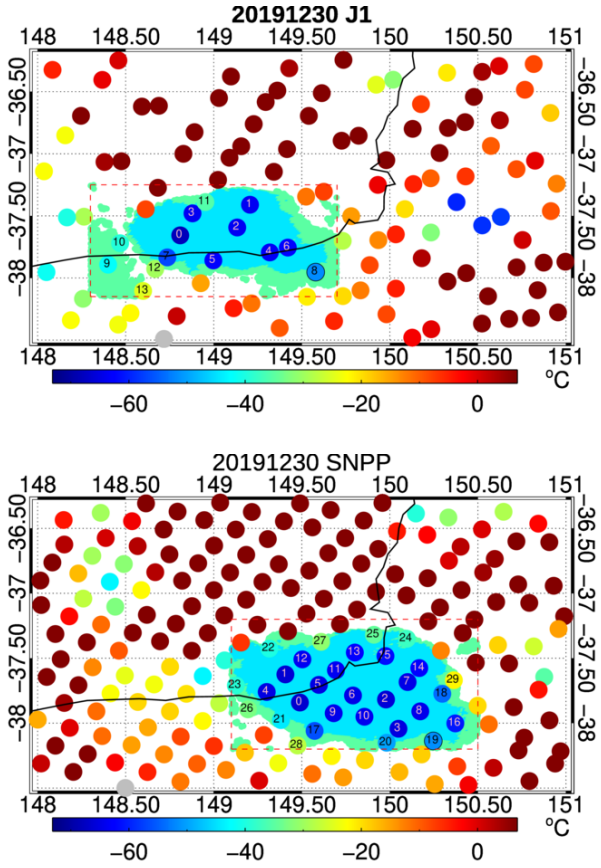


Fig. 2. BT at 11 μm for CrIS overlaid with (top) VIIRS from J-1 (14:50 UTC) and (bottom) S-NPP (15:12 UTC) on December 30, 2019. The pyroCb is highlighted with the squares (marked with red dashed lines). Only the pixels within the box with BT11 < −20 °C are numbered, starting from the lowest CrIS BT11. Color bars represent CrIS BT11 values, while VIIRS data are displayed in two colors: light blue for pixels with VIIRS BT11 < −60 °C and light green for the pixels with VIIRS BT11 between −35 °C and −60 °C. Each circle marks the location of CrIS pixels but does not reflect the actual CrIS pixel size.

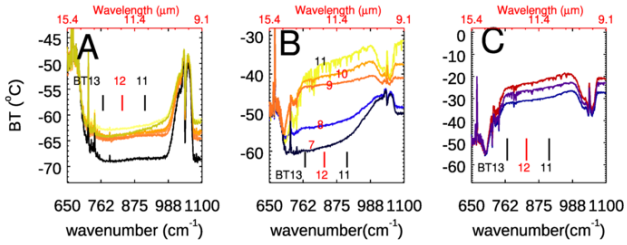


Fig. 3. Spectra for 15 CrIS pixels of J-1 (see Fig. 2), plotted in three groups with BT11 < −60 °C (A), −60 °C~35 °C (B), and −35 °C~−20 °C (C). Different colors are used to represent different spectra for visual clarity. The pixel numbers #0–#6 in group (A) and #12–#14 in group (C) are not shown. Only the pixel numbers #7–#11 in group (B) are depicted in the middle. Please note that the y-scale ranges are different.

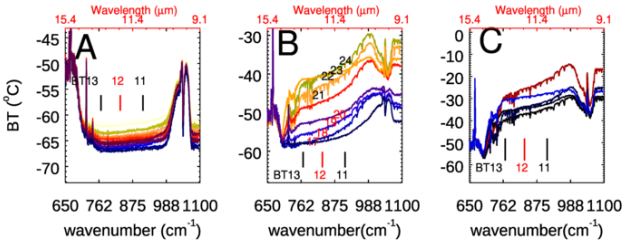


Fig. 4. Spectra for 31 CrIS pixels of S-NPP (see Fig. 2), plotted in three groups with BT11 < −60 °C (A), −60 °C~−35 °C (B), and 35 °C~−20 °C (C). Different colors are used to represent different spectra for visual clarity. The pixel numbers #0–#16 in group (A) and #26–#30 in group (C) are not shown, and only #17–#24 in group (B) are displayed in the middle.

cloud drops that mostly freeze homogeneously and suppress precipitation-forming processes [10]. In addition, the pyroCb cloud-top BTs in the long-wave IR channel near 11 μm (BT11) are usually below an approximated homogeneous liquid-water freezing threshold (i.e., BT11 < −38 °C) [10]. A study on the pyroCb during 2019–2020 Australia fires [17] showed that the most intense blowup fires and the highest pyroCb echo-top altitudes (via subevents #11 and #12) were observed over Victoria at ~16:00 UTC. This pyroCb can be illustrated from the BT11 for one granule of VIIRS onboard J-1 (~14:24 UTC) and another one of VIIRS onboard S-NPP (~15:12 UTC) (Fig. 1). The BT11 in the region marked with blue color is mostly less than 213.75 K (−60 °C), which is the region covered by pyroCb. From the size of the pyroCb in Fig. 1 (blue color area within the box), we can see the expansion of the pyroCb anvil area. Based on these two VIIRS snapshots in 50 min apart, this pyroCb moved eastward by ~0.5° (which is equivalent to ~50 km), and the area of the pyroCb has almost doubled.

Fig. 2 plots the corresponding BT from the CrIS channel at 11.1 μm (BT11). For later analysis in this article, we assign a number to each CrIS pixel if its BT11 is less than −20 °C and located within the spatially coincident box (see Fig. 1). The sequencing of numbers is assigned based on the increase of BT11, with the number "0" assigned for the pixel with the lowest BT11. Since CrIS has much coarser spatial resolution than VIIRS, the higher resolution VIIRS pixels are overlaid in two shaded colors to show whether a CrIS pixel lies within the pyroCb core (BT11 < −60 °C) or in its periphery (−60 °C < BT11 < −35 °C). Since the pyroCb cloud-top temperature is usually below the approximate homogeneous liquid-water freezing threshold of −38 °C, these central pixels with BT11 < −60 °C, shown in dark blue, can be confidently identified as pyroCb pixels. It is evident from Fig. 2 that the number of these pyroCb pixels is from 0 to 6 at 14:24 UTC (from J-1) and from 0 to 17 at 15:12 UTC (from S-NPP), respectively. As shown in Fig. 1, the eastern front edge of the pyroCb anvil, as illustrated from the shaded area of VIIRS, is located at ~149.75°E (longitude) from J-1 and ~150.5°E (longitude) from S-NPP, respectively, demonstrating the pyroCb anvil expanded by ~75 km toward the east. At the same time, the expansion of the pyroCb anvil to the south is about 0.2° (~20 km) within ~50 min. The expansion of the pyroCb toward the ocean in the southeast of Australia has also been verified from the wind fields (not shown). This rapid development of pyroCb, as shown in Figs. 1 and 2, is consistent with the finding that the most intense blowup of pyroCb was recorded around 16:00 UTC [17].

The special characteristics of pyroCb IR spectra can be seen from Figs. 3 and 4 that plot all the spectra of the pixels numbered within the box (Fig. 2). These pixels are

put in three groups based on BT11 as: BT11 < −60 °C (A), −60 ~ −35 °C (B), and > −35 °C (C). Since a very cold cloud temperature of 210 K can be used to identify bright uniform deep convective clouds (DCCs) (e.g., [38]), group (A) represents intense pyroCb pixels. Later, we will show that group (C) includes some non-pyroCb pixels, and group (B) consists of some low-level pyroCb pixels and some pixels in semi-transparent pyroCb-blowoff periphery. From Figs. 3 and 4, it is evident that the intense pyroCb spectra in group (A) exhibit a peak near 9.6 μm, whereas the pixels in group (C) display a valley. These pixels in group (A) with very low BT11 (less than −60 °C) are likely associated with intense updraft, a phenomenon that has been found in [17]. The spectra for pixels in group (A) have a few common features: 1) the difference among BT11, BT12, and BT13 is very small, suggesting that they are optically thick cloud; 2) the spectra from 760 cm^{-1} (13.2 μm) to 980 cm^{-1} (10.2 μm) are very smooth, and the feature of water vapor line absorption is almost unnoticeable; and 3) an inverted "V" shape near the O_3 absorption band at 9.6 μm. As the temperature decreases with altitude and reaches the minimum in the tropopause, if the cloud top is near or above the tropopause, the peak O_3 absorption channels near 9.6 μm receive additional emission from higher, warmer atmospheric layers, leading to higher BTs for these peak absorption channels and the inverted "V" spectral feature 3) around 9.6 μm. Please note that inverted spectra are not a unique feature exclusive to pyroCb events. This phenomenon has been widely recognized as evidence of overshooting convection, as documented in previous studies (e.g., [41], [61]). The inverted "V" feature near 9.6 μm for pyroCb is opposite to the common "V" shape for clear sky or low clouds, as shown for the pixels in the group (C). Later, in this article, we will show that these pixels in group (C) have much lower cloud-top height and smaller optical depth than pixels in group (A).

The smooth feature 2) in group (A) may result from a strong cloud signal overwhelming the water vapor signature. Fewer absorption lines also indicate these pixels are above a high cloud where the water vapor is very low. One common spectral feature in IR for clear sky or over low water clouds is the prominent water absorption lines that appear as numerous spikes throughout the spectral region between 760 and 980 cm^{-1} (or 13.2–10.2 μm). More details of this water vapor absorption feature can be found in [62] and [63]. These line structures are evident in the spectra of pixel #11 in group (B) and all pixels in group (C). Later, in this article, we will show pixel #11 is likely a partial pyroCb anvil.

Since BT11 < −35 °C has been used to identify intense pyroCb (e.g., [25]), we will focus on the detection of these pyroCb pixels in group (B), where their BT11 is between −35 °C and −60 °C, and some of them have inverted "V" spectral features.

The split-window technique has been widely used for the detection of cirrus and blackbody clouds (e.g., [64]) and cloud-top height estimation [65]. It is also used to detect pyroCb (i.e., [25], [26]). For instance, both an RTM and satellite observations indicate that BT11 (910 cm^{-1}), BT12 (833 cm^{-1}), or BTD11–12 are near zero for clear sky (warm thermal IR) and optically thick water and ice clouds [66], while a large BTD11–12 of ~ +3 K is found for cirrus cloud with nominal effective radius (~ 30 μm). Peterson et al. [25] also found that the BTD11–12 is near 0 °C at the warm (clear air) and cold (optically thick cloud) extremes of each scene cluster of Chisholm pyroCb, but the arcing BTD11–12 in between these limits for Phases I and II plume segments of Chisholm pyroCb reached maximum values of roughly +6 °C and +9 °C, respectively (Fromm et al. [26], Fig. 3). For all pixels in group (A), BTD11–12 is near zero, suggesting they are optically thick, cold ice clouds, while BTD11–12 are mostly between 1 °C and 2 °C for pixels in group (B), suggesting that they are not over thick intense ice clouds and may have some small ice particles.

Since CrIS has many channels, we applied a linear fitting to the spectra in the region of 800–980 cm^{-1} (12.5–10.2 μm) to determine the slope. Previous studies have shown that a large slope value is linked to the presence of small ice particles within the cloud. For instance, using aircraft measurements, Smith et al. [67] found that for clouds consisting of highly absorbing water droplets or large ice crystals, the BTs within the 8–12 μm (1250–833 cm^{-1}) region is nearly constant, but for clouds with a nearly invisible layer of small ice particles, and there is a very strong spectral feature between 10 and 12 μm (1000–833 cm^{-1}). Model simulation by Yang et al. [68] demonstrated that the BTs of cirrus clouds vary significantly with wavenumber between 750 and 1000 cm^{-1} (13.3–10.0 μm). For small ice particles, the BTs from 800 to 1000 cm^{-1} (12.5–10.0 μm) increase rapidly with the wavenumber, and the amplitude of the slope within the wavelength range increases as ice particle size decreases. Therefore, a larger slope value indicates a cloud with small ice particles. In addition to using the fitting slope, we also calculate the STDs, as a lower STD value indicates smaller variation in BT, corresponding to a smoother spectrum with fewer line structures linked to water vapor absorption. Less water vapor absorption suggested a drier air condition above the cloud layer in the higher atmosphere. To calculate the STD, two separate linear fittings were performed: one in 800–900 cm^{-1} (12.5–11.1 μm) and another one in 900–980 cm^{-1} (11.1–10.2 μm), and the fitting errors from these two spectral regions were used together.

From Fig. 5, we can see that pixels in group (A) have smaller slopes (<0.02) and STD (<0.2), while the variations for pixels in groups (B) and (C) are large and cannot be distinghed between these two groups. Pixels #9–#11 from J-1 and pixels #21–#24 from S-NPP have a regular "V" shape near 9.6 μm. The regular "V" shapes, combined with these large slope and STD values, suggest that these pyroCb pixels are likely at a lower pressure level (i.e., below ~200 hPa) but contain small ice particles. In contrast, pixels with smaller slope and STD values indicate they might be at relatively higher altitudes or in drier atmospheric conditions. The larger slopes for the pixels in group (B) may result from the presence of smaller ice particles within these pixels and/or their position at the edge of the pyroCb anvil (see Fig. 2). A larger slope also corresponds to a greater value of BT11–BT13. It was found in [25] that for some partial clouds or thin clouds passing over a hot fire, a test of BT11–BT13 > 3 K can be used to

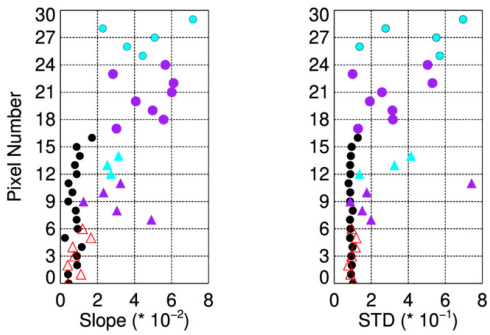


Fig. 5. Slopes of a linear fitting in the spectral regions (760–980 cm^{-1}) (or 13.2–10.2 μm) and the corresponding STD for CrIS pixels. For J-1, red triangles are for pixels #0–#6 in group (A), solid purple triangles for pixels #7–#11 in group (B), and solid cyan triangles for pixels in group (C). For S-NPP, solid black circles are for pixels #0–#16 in group (A), solid purple circles for pixels #17–#24 in group (B), and solid cyan circles for pixels in group (C). The locations of pixel numbers are shown in Fig. 2.

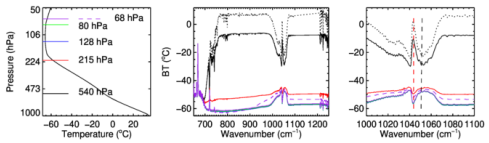


Fig. 6. (Left) Temperature profile from ERA5 and (right two) the PCRTM-simulated spectra. Different colors for the spectra correspond to clouds with different top heights marked with the same color on the left. Dark dashed line on the right is for clear sky, and the pink dashed line is for ice cloud with a smaller effective radius of De = 30 μm. The right provides a zoomed-in view of the spectrum in 1000–1100 cm^{-1}, with vertical red and black dashed lines indicating the positions of 1043.75 and 1051.645 cm^{-1}, respectively.

reduce potential cloud edge noise and detection errors from thin clouds passing over a hot fire. From a visual examination in Fig. 2, we can see that pixels #7 and #8 from J-1 and #17–#21 from S-NPP are indeed located in the boundary of dark blue and light blue areas and are likely at the cloud edge. While BT11 = −35 °C has been used as a threshold of BT11 for the detection of ice particles in pyroCb using sensors with broadband channels (e.g., [17]), our analysis of pyroCb detection for pixels in group (B) suggests that a simple use of BT11 < −35 °C for intense pyroCb detection is inappropriate for hyperspectral IR sensors; instead, it is more reliable to use the entire long-wave band spectral information, such as the slope, STD, and spectral shape near 9.6 μm.

B. CrIS Spectra From RTM Simulation and Its Use for PyroCb Detection

To better assess the influence of clouds on the spectra, we used the PCRTM model to simulate CrIS observations. In addition to cloud properties, the model inputs include temperature, water vapor, and ozone profiles from the ERA-5, matched with the CrIS observation at the center of the analysis box (Fig. 1). The temperature profile is shown in Fig. 6, with the cloud tops indicated by different horizontal lines. For the cloud top at 540 hPa, we assume a water cloud with an effective particle diameter (De) of 12 μm, and all higher level clouds are treated as a single-layer ice cloud with

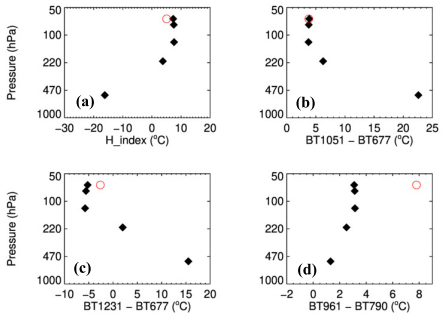


Fig. 7. Variation of (a) H_index, (b) BT1051–BT677, (c) BT1231–BT677, and (d) BT961–BT790 for clouds with tops at different pressure altitudes based on PCRTM simulations. Red dots indicate clouds at 68 hPa but with a smaller particle size.

De = 48 μm, except for the cloud at 68 hPa, where the De with a small values is also simulated (the pink dashed line corresponds to De = 30 μm, with its spectra presented later). For all cases, the cloud optical depth is fixed at 6, which is sufficiently large based on our retrieved cloud properties. The surface emissivity is assumed to be 0.97.

Fig. 6 clearly shows that, as expected, the spectra for low water clouds closely resemble those of clear sky (dark dashed line), with both exhibiting a pronounced "V" feature near 9.6 μm. With the cloud tops moving up to a higher altitude of about 200–250 hPa, the spectra become inverted, and the inverted "V" becomes deeper as the cloud top moves closer to the tropopause. It is evident from Fig. 6 that the spectra become inverted for clouds with tops at 215 hPa or higher. The peak height of the inverted "V" is relatively small for the cloud with a top at 215 hPa, and it increases with the cloud top increase. However, the spectra for clouds with the tops at 128, 80, and 68 hPa almost overlap with each other, and this is because the temperature in these three levels is almost the same. For ice clouds at the same cloud-top height of 68 hPa but having a smaller particle size (De), BT has a significant increase between 750 and 1000 cm^{-1} (or 13.3–10.0 μm), as shown in the pink dashed line.

For convenience in later analysis, here, we introduce a new variable called "height index" (H_index). It is defined as the difference between the maximum and minimum BT within the spectra range of 995–1065 cm^{-1} (or 10.1–9.4 μm). The maximum BT occurs at 1050.625 cm^{-1} (9.52 μm) and is designated as BT1051, while the minimum BT is located near the center of this spectral range at 1043.75 cm^{-1} (9.58 μm), which is the most transparent channel in the ozone band near 9.6 μm. For noninverted spectra over low water clouds or clear skies, BT1051 is smaller than BT at 1043.75 cm^{-1}, resulting in an H_index of less than 0. Since the BT at 1043.75 cm^{-1} is close to the BT in the window channels between 10 and 14 μm and approximately represents the cloud-top temperature for thick clouds, with the lowest value at the tropopause, the H_index reaches its maximum when the cloud top is near the tropopause. Fig. 7(a) illustrates the variation of H_index with different cloud-top heights using the simulated data. As shown in Fig. 7(a), H_index is greater than 5 for all high clouds and reaches approximately 10 for cloud tops near 100 hPa, making

it an effective parameter for detecting high clouds. However, for clouds with tops at 128, 80, and 68 hPa, the difference in H_index is small, making it difficult to distinguish between them based solely on H_index. In such cases, some other split-window detection approaches will be required, as outlined in the following.

Like the DCC test, specific channels can be selected from CrIS for the pyroCb test. Aumann et al. [41] used the BT of the AIRS channel at 1231 cm^{-1} (8.1 μm) minus BT at 712 cm^{-1} (14 μm) to detect DCCs and used the BT at 961 cm^{-1} (10.4 μm) minus BT at 790 cm^{-1} (12.6 μm) to discriminate clouds above or below the tropopause. For simplicity, the BT at 1231 cm^{-1} (8.1 μm), 961 cm^{-1} (10.4 μm), and 790 cm^{-1} (12.6 μm) will be referred to as BT1231, BT961, and BT790, respectively. We will adopt these two tests from Aumann et al. [41] but with a modified CrIS stratospheric channel at 676.875 cm^{-1} (14.8 μm), which is designated as BT677, instead of the channel of AIRS at 712 cm^{-1} [41]. Based on the temperature Jacobian, the peak sensitivity of the channel at 712 cm^{-1} corresponds to a higher altitude near 10 hPa, while the peak sensitivity of the channel at 676.875 cm^{-1} is near 50 hPa. The average BT677 in the region near the pyroCb, indicated by the square in Fig. 2, is −50.35 °C ± 0.20 °C for J-1 and −52.10 °C ± 0.19 °C for S-NPP. In comparison, the average BT at 712 cm^{-1} is −56.82 °C ± 5.94 °C for J-1 and −57.95 °C ± 5.50 °C for S-NPP. Given that the variability of BT677 is smaller than that at 712 cm^{-1} in both the J-1 and S-NPP scenes, BT677 is selected as the reference for estimating the pyroCb cloud-top height using CrIS data. It is evident that BT1051–BT677 [Fig. 7(b)] effectively differentiates high clouds above the tropopause from those significantly below it. A BT1051–BT677 value greater than 10 indicates low cloud. From the variation of BT1231–BT677 and BT961–BT790 for different cloud tops using the simulation data, as shown in Fig. 7(c) and (d), it is evident that the clouds below the tropopause, i.e., at 215 hPa and water cloud at 540 hPa, can be separated from high clouds based on BT1231–BT677 [Fig. 7(c)], and the cloud with small ice particles (red dot) can be separated from other clouds using BT961–BT790 [Fig. 7(d)]. A large BT961–BT790 value greater than 4 can be used to discriminate clouds with smaller ice particles.

Fig. 8 is like Fig. 7 but shows the variation of H_index, BT1051–BT677, BT1231–BT677, and BT961–BT790 for pixels using actual CrIS observation data from J-1 and S-NPP on December 30, 2019. It is evident from Fig. 8 that for all pyroCb pixels in group (A) (#0–#6 from J-1 and #0–#16 from S-NPP) share the same characteristics: H_index ≅ 10 K, BT1051–BT677 < 5 K, BT1231–BT677 < −5 K, and BT961–BT790 < 2 K. These parameters overall agree well with those high clouds from model simulations, and some small discrepancy is primarily due to differences in cloud optical properties, emissivity, temperature, and water vapor profile used in the PCRTM. For pixels in group (B), the values of H_index for pixels #7 and #8 from J-1 and pixels #17–#19 from S-NPP are in a range of 4–7, BT1231–BT677 in a range of 10–15, and BT1051–BT677 in a range of 16–20. These values are similar to those from simulations with a cloud top at 215 hPa, suggesting that these pyroCb pixels should be at

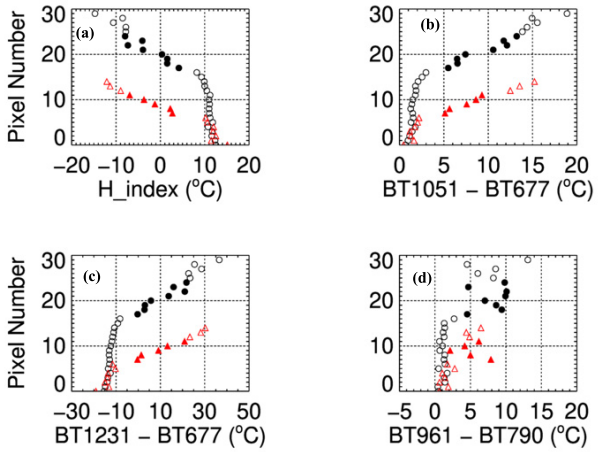


Fig. 8. Variation of (a) H_index, (b) BT1051–BT677, (c) BT1231–BT677, and (d) BT961–BT790 for pixels of J-1 (red triangles) and S-NPP (dark circles). Solid red triangles (#7–#11) and solid dark circles (#17–#24) are for pixels in group (B) of J-1 and S-NPP, respectively. The locations of pixel numbers are shown in Fig. 2.

relatively lower altitudes of ~215 hPa below the tropopause. A large negative value of H_index for pixels #10 and #11 from J-1 and pixels #22–#25 from S-NPP, along with the obvious water vapor line absorption features in the spectra from 750 to 990 cm^{-1} (or 13.3–10.1 μm) and deep "V" shapes near 9.6 μm (see Figs. 3 and 4), indicates that these are low water clouds or partial clouds. Given that their H_index values are larger than those from the simulation with a cloud top at 540 hPa, they are likely situated above 500 hPa. The values of BT961–BT790 for pixels #7 from J-1 and #18–#25 (except 23) are similar to those from simulation with a small ice particle size, suggesting that they likely consist of small ice particles.

C. PyroCb Cloud-Top Height From SiFSAP Retrieval Products

While hyperspectral radiance information can be used for pyroCb detection, as discussed before, a more accurate determination of the cloud-top height can be achieved from the retrieval products like SiFSAP. This is because these products are derived from the entire spectral dataset, ensuring radiance closure during the retrieval process, and have been validated [33]. Although some uncertainty could arise as the cloud LUTs used in the retrieval do not account for the special complexity of pyroCb, particularly the mixture of fire particle aerosols, the retrieved products from SiFSAP are still expected to provide valuable insights into this pyroCb. Fig. 9 shows the SiFSAP-retrieved cloud-top pressure and optical depth for all the pixels in groups (A)–(C) from J-1 and S-NPP. As shown in Fig. 9, the cloud tops for pixels #0–#6 from J-1 and #0–#16 from S-NPP in group (A) are mostly above the tropopause, with the mean cloud-top height of 14.53 ± 2.43 km and 15.41 ± 2.34 km, respectively. The mean retrieved optical depth for the seven pixels (#0–#6) from J-1 is 5.45 ± 0.81, which is smaller than the mean optical depth of 7.46 ± 1.09 for the 18 pixels (#0–#17) from S-NPP. This increase in the mean cloud optical depth, along with the increase in the number of pyroCb pixels from 7 to 18, highlights the rapid development of this pyroCb

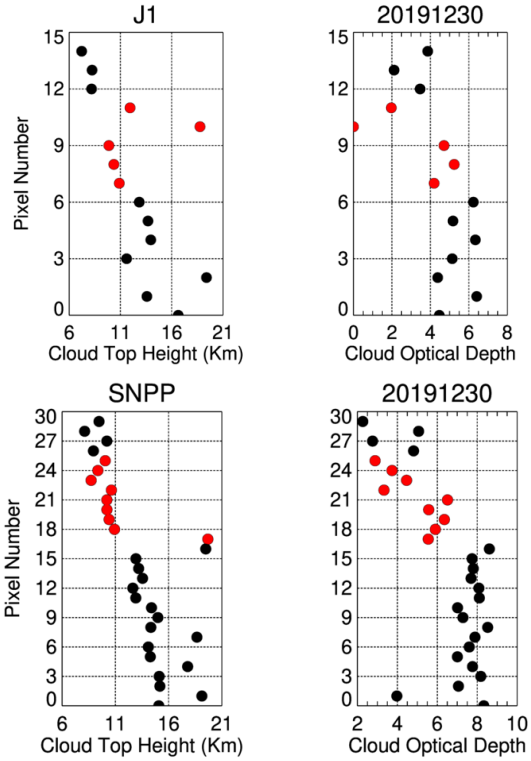


Fig. 9. SiFSAP cloud-top height and optical depth, retrieved using CrIS/ATMS on J-1 (14:20 UTC) and S-NPP (15:12 UTC). Solid red dots mark the pixels from group (B) (see Fig. 3).

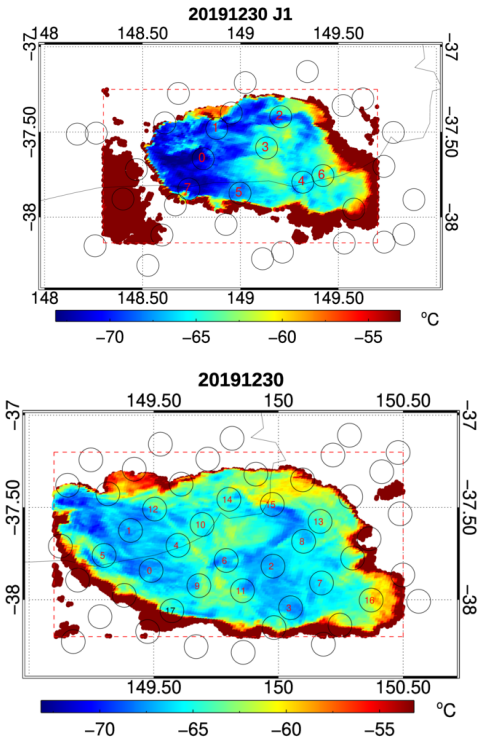


Fig. 10. Map of VIIRS pixels with BT11 < −35 °C, overlapped with the location of CrIS pixels from J-1 and S-NPP. Only the numbers for pixels in group (A), except pixel #17 from S-NPP, are labeled. Unlike Fig. 2, the color bar represents VIIRS data only, with a narrow range to highlight the cloud features surrounding each CrIS pixel. Each circle marks the location of CrIS pixels but does not reflect the actual CrIS pixel size.

and expansion of the cloud anvil from 14:24 UTC (J-1) to 15:12 UTC (S-NPP). The mean cloud-top height and the mean optical depth for pixels in group (C) are 9.17 ± 0.68 km and 3.42 ± 1.25, which are lower than 11.13 ± 3.09 km and 4.92 ± 1.28 from group (B), respectively.

Using SiFSAP-retrieved temperature profiles, we calculate the thermal tropopause height using the method from Reichler et al. [60]. The mean tropopause near the pyroCb region is about 164.8 hPa for the S-NPP observation on 15:12 UTC, which is approximately 13.0 km. Using the SiFSAP-retrieved cloud-top pressure, the relative height of the cloud top versus the tropopause can be estimated. As shown in Fig. 9, there are seven pyroCb pixels (#0 and 2#–#6) from J-1 and 18 pyroCb pixels (#0–#17) from S-NPP, shooting up near or above the tropopause. Notably, pixels #0 and #2 from J-1 and #1, #5, #8, #16, and #17 from S-NPP are 3–6 km above the tropopause. Further examination of the spectra discrepancy between the observations and the fit ones in the retrieval, an indicator of retrieval quality, reveals that the discrepancy for pixel #1 (S-NPP) is relatively large and should be excluded. By conducting a zoomed-in view of visual inspection of the VIIRS observation matched with CrIS data (see Fig. 10), it is clear that pixel #17 samples the semi-transparent pyroCb-blowoff periphery, and there is a large temperature gradient surrounding it. While the retrieved high cloud-top heights for pixels #1 and #17 from S-NPP are unreliable, the elevated plume heights for pixels #5, #8, and #16 from S-NPP and pixels #0 and #2 from J-1 do seem reasonable. Using data from GOES-17, it was found in [17] that the combined anvil cloud (from subevents #11 and #12) reached altitudes of at least 15.8 km (BT11 < −70 °C), exceeding the tropopause by 2.3 km in the middle of the night on 30 December (~16:00 UTC, 03:00 LT). From our study, the mean altitude of pixels #0–#17 from S-NPP at 15:12 UTC is 2.41 ± 2.34 km above the tropopause of 13.0 km, which generally agrees with the results from [17]. However, according to SiFSAP retrievals, the cloud-top heights are 16.6 and 19.4 km for pixels #0 and #2 from J-1, and 17.8, 18.7, and 19.5 km for pixels #5, #8, and #16 from S-NPP, respectively. Using the same tropopause height of 13.5 km from [17], these cloud-top heights exceed the tropopause by 3–6 km, suggesting a potentially higher plume height from 2019 to 2020 Australian fires than previously reported. It should be noted that the presence of smoke may significantly alter the radiative properties of the clouds, introducing uncertainty in the retrieved cloud-top height. Therefore, the difference between our results and those from [17] could be attributed to the different instruments and methods used in estimating cloud-top height. Further improvements in the retrieval accuracy could be achieved with a more accurate pyroCb fire-cloud model. A visual examination of the fine structure of VIIRS BT11 around these pixels (see Fig. 10) suggests that the high plumes observed for three pixels over land (i.e., pixels #0 and #2 from J-1 and #5 from S-NPP) are likely due to ejection from newly generated "young" smoke plumes, while the high plumes for

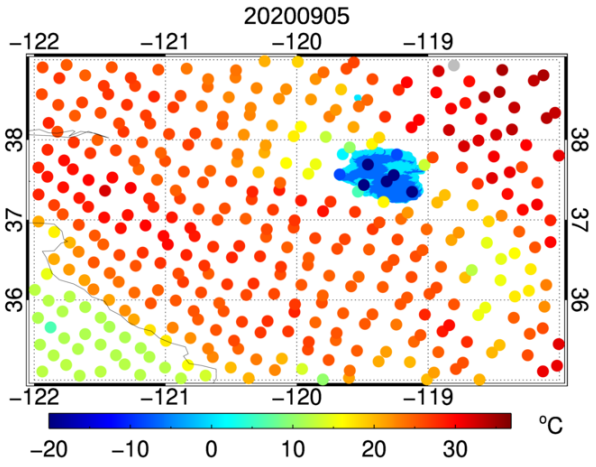


Fig. 11. BT11 for CrIS and the matchup of VIIRS on S-NPP (21:18 UTC, September 05, 2020). Color bar represents CrIS BT11, while the VIIRS data are marked using two colors: dark blue for pixels with BT11 < −35 °C and light blue for pixels with BT11 between −35 °C and −20 °C.

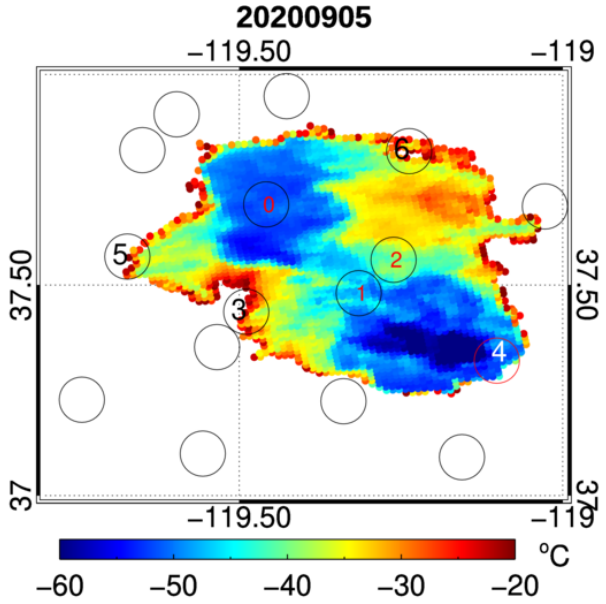


Fig. 12. PyroCb mapped with VIIRS pixels with BT11 < −20 °C. The locations of CrIS pixels with BT11 < 0 °C are overlaid. These CrIS pixels cover some of the pyroCb and are numbered according to the temperature of CrIS BT11, and #0 is for the pixel with the lowest BT11. The size of each circle does not represent the real pixel size of CrIS, which is about 14 km in diameter for a pixel at nadir.

the two other pixels over the ocean (i.e., #8 and #16 from S-NPP) are more likely due to pyroCb expansion.

D. Evaluation of PyroCb Detection and Cloud-Top Height Estimation From 2020 California Creek Fires

From RTM simulation results, Aumann et al. [41] demonstrated that a temperature difference of DT (DT = BT1231–BT712) < −2 K effectively distinguishes cloud tops located more than 5 hPa below the tropopause from higher altitude clouds, and DC (DC = BT961–BT790) >0.5 K can be used to separate clouds that are over 5 hPa above the tropopause from those more than 5 hPa below it. Using SiFSAP cloud-top pressure and pyroCb pixels from Australian fires, we can apply similar criteria as follows: 1) DT >0 K for pyroCb pixels below the tropopause; 2) DC <1.2 K and DT < −2 K for pyroCb pixels more than 5 hPa above the tropopause. Even more, we can have criteria 3) H_index >0 for pyroCb pixels above 250 hPa; and 4) H_index >10 and BT1051–BT677 < 1.5 k for pyroCb pixels above the tropopause.

To validate the spectral features of pyroCb discussed in Sections III-A and III-B and to explore the application of these hyperspectral characteristics for pyroCb detection and cloud-top height estimation, we analyzed the 2020 Creek Fire, which occurred in central California's Sierra National Forest in Fresno and Madera counties. This large wildfire began on September 4, 2020, driven by up-canyon winds and a thermodynamic profile conducive to moist convection aloft [69]. On September 5, the pyroCb was observed using radar and satellite data. GOES-17 true-color imagery displayed a broad smoke plume with distinct white pyroCb cloud tops, and NEXRAD radar data showed that the pyroCb plume contained an updraft core with overshooting plume tops reaching approximately 15.5-km MSL and a broader shield of lower plume/cloud tops extending northward from the fire (Fig. 1, [70]).

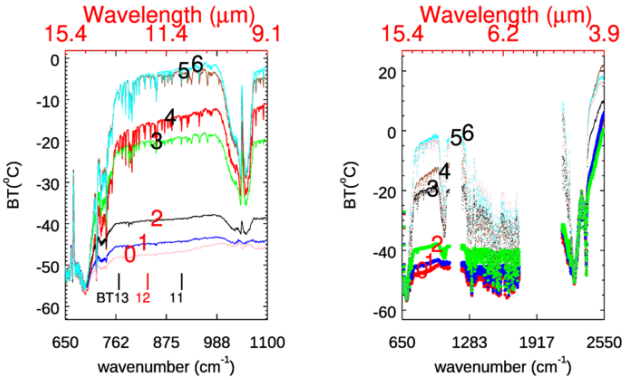


Fig. 13. (Left) Spectra for CrIS pixels #0–#6 (see Fig. 12) in the long-wave band and (right) the whole spectral range. Different colors are used to represent different spectra for visualization.

The pyroCb on September 5 was captured by VIIRS and CrIS on S-NPP, as shown in Fig. 11. By applying the same threshold used in [25], the deep convection area can be identified using VIIRS BT11 < −20 °C, which is highlighted in the blue area near the center of Fig. 11. A zoomed-in view of this area, as shown in Fig. 12, reveals that seven CrIS pixels, labeled #0–#6, cover the pyroCb. The BT11 from CrIS is less than −35 °C for pixels #0–#2, indicating that they are DCCs with ice particles. Fig. 13 plots the observed spectra for these seven CrIS pixels. Since this pyroCb was observed during the daytime, the spectra near 4 μm include contributions from both emitted thermal radiation and reflected solar radiation. Small cloud droplets and ice particles create relatively high solar reflectivity, leading to a higher BT in the short-wave channels near 3.9 μm during the daytime, as

TABLE I
LIST OF CHARACTERISTIC PARAMETERS FOR PYROCB PIXELS FROM CALIFORNIA CREEK FIRE ON SEPTEMBER 5, 2020

Pixel #	Slope	STD	H_index	BT1051 – BT677 (K)	BT1231 – BT677 (K)	BT961 - BT790(K)	SiFSAP Cloud-Top Pressure (hPa)	SiFSAP Optical Depth	BT2550-BT909(K)
0	0.014	0.11	1.08	7.95	7.16	2.66	219.4	3.5	52.48
1	0.010	0.12	0.60	8.21	8.84	1.83	223.2	4.8	50.41
2	0.009	0.10	-2.73	10.62	14.50	1.82	261.6	4.6	39.05

shown from the whole spectra in Fig. 13. Thus, following the approach used in [25], the difference in BT between 3.9 μm and the window channel can be effectively utilized for pyroCb detection during daytime. To examine the spectral shape near the 9.6-μm O_3 band, a portion of the long-wave spectra is plotted as done in Figs. 3 and 4. Among these seven pixels, there are a lot of spikes in the long-wave spectra between 760 and 980 cm^{-1} (or 13.2–10.2 μm) for pixels #3–#6, which represent water absorption lines. This characteristic, along with the pronounced "V" shape near the 9.6-μm O_3 band, indicates that these four pixels are possibly over low clouds, clear areas, or partial clouds. A visual inspection of Fig. 12 reveals that pixels #3–#6 are indeed along the edge of the pyroCb, suggesting that they are likely partially cloud-covered. In addition, the CrIS BT11 for pixels #5 and #6 is above −10 °C, further confirming that they are not DCCs as they failed to pass the deep convection test of −20 °C [25]. The lobe of cold cloud under pixel #0 was generated by a pyroCb pulse about an hour before S-NPP sampling. The active pyroconvection was located closer to pixel #4, where a cold lobe formed around 21:00 UTC and remained active at the time of S-NPP observation. The blowoff direction was toward the north–northwest [69], [70]. Even though pixel #4 is located near the most intense pyroCb plume as depicted by VIIRS, its BT11 is −19.0 °C, indicating that it is only partially covered by pyroCb. In this case, applying a simple threshold of BT11 < −35 °C identified 2625 VIIRS pixels, whereas only three CrIS pixels fully captured the pyroCb. This highlights a limitation of current hyperspectral sounders for pyroCb studies.

Although the spectrum of pixel #2 shows a distinct "V" shape near the 9.6-μm O_3 band, its depth is shallower than that of pixels #3–#6, suggesting that pixel #2 is at a higher altitude than #3–#6. The smoothness of the spectrum in the 760–980-cm^{-1} range suggests reduced water line absorption, indicating drier atmospheric conditions. A visual inspection in Fig. 12 confirms that part of pixel #2 is over intense pyroCb (BT11 < −35 °C) and part of it is over marginal pyroCb, as defined by BT11 between −35 °C and −25 °C [25]. The inverted "V" spectra appear for CrIS pixels #0 and #1, but with shallow depth. This feature suggests that pixels #0 and #1 are at even higher altitudes than pixel #2. Like pixel #2, the spectra for these pixels are very smooth between 760 and 980 cm^{-1}, indicating that they are also in a dry atmospheric condition. The actual altitudes for these three pixels retrieved from SiFSAP are listed in Table I. These values are consistent with previous analysis based solely on spectral information. In addition, Table I includes other relevant parameters, as previously discussed for Australian pyroCb, for pixels #0–#2, as well as the values of BT4–BT11 that can be used for the daytime pyroCb test (e.g., [25]). For pixels #0 and #1, BT4–BT11 are greater than 50 °C, confirming that they are pyroCb pixels. However, pixel #2 did not pass this pyroCb test as the BTD4-11 is less than 50 °C, underscoring the discrepancies in thresholds when using broadband versus hyperspectral channels for pyroCb detection. Comparing the parameters in Table I with Figs. 5 and 8, the values of H_index, BT1051–BT677, and BT1231–BT677 for these three pixels from California Creek Fire are mostly similar to those of pixels #7–#9 (J-1) and #17–#20 (S-NPP) from Austrlian fires, suggesting that these parameters are typical for pyroCb at altitudes slightly below 200 hPa. However, the slope and STD values, along with BT961–BT790 for these three pixels from California Creek Fire, closely resemble those of higher altitude pyroCb pixels in group (A) (#0–#6 from J-1 and #0–#16 from S-NPP) from Australian fires. This similarity indicates an even drier atmospheric condition for California pyroCb than Australian pyroCb.

To determine if the above tests and characteristic parameters for pyroCb can effectively distinguish them from regular Cb clouds, we compared these parameters between pyroCb pixels and other Cb pixels within the same CrIS granule from J-1 on December 30, 2019. The Cb pixels are visible in the BT11 as dark blue in the upper panel of Fig. 14 (for J-1 only), while the lower panel shows the locations of pixels with high cloud tops. A comparison of these characteristic parameters between pyroCb pixels and Cb pixels is illustrated in Fig. 15. The pyroCb pixels are those from groups (A) and (B), specifically pixels #0–#11of J-1 and #0–#25 of S-NPP, all exhibiting BT11 values below −35 °C. Considering the clumps of cold cloud near and off the coast are likely the blowoff from other pyroCb pulses, we have selected only the Cbs pixels south of –42°S with BT11 values lower than −35 °C for comparison. From Fig. 15(a) and (b), it is evident that for most pyroCb pixels, the differences in these four parameters between pyroCb and Cb pixels are minimal, indicating that it is still challenging to use hyperspectral IR only to distinguish pyroCb from Cbs during nighttime. We found a combined use of BT1231–BT677 and H_index is promising in distinguishing these high pyroCb in group (A) from other Cb pixels, as illustrated in Fig. 16. Since pyroCb is always linked to extreme fires, we can use other observations,

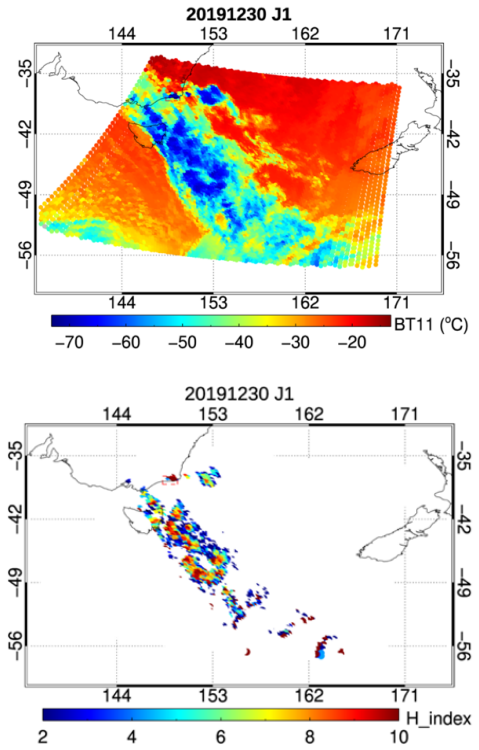


Fig. 14. (Top) Map of CrIS BT11 from J-1 on December 30, 2019, (bottom) along with the corresponding H_index. The largest H_index values correspond to pixels with cloud-top heights near or above the tropopause.

such as imager, to assist in the identification of fire locations. Thus, using hyperspectral IR spectral information and SiFSAP products can further improve the observation of pyroCb and its cloud-top heights.

IV. Summary and Conclusion Remarks

Large wildfires and their induced pyroCb are getting increasing interest and attention, as the wildfire-induced pollutants and smoke pose great risks to air quality and human health, and pyroCb provide an efficient chimney-like smoke pathway for the transport of biomass-burning aerosols, gases, and cloud ice particles to the stratosphere. Most previous studies were based on the observations from broadband satellite sensors; however, the potential benefits of using hyperspectral IR sounders for pyroCb study have not been fully explored even though there are over 20 years of hyperspectral observation data from instruments like AIRS on EOS Aqua (since 2002), IR Atmospheric Sounding Interferometer (IASI) on European Space Agency (ESA) MetOp satellites (since 2006) [71], and CrIS (since 2011). On the basis of previous studies that found exceptionally high pyroCb activity caused by Australia's unprecedented fires on the evening of December 30, 2019, this research utilized the unique CrIS measurement data from JPSS-1 and S-NPP to explore the potential of hyperspectral information, along with the SFOV retrieval products, for pyroCb detection and plume height estimation. The findings were further validated with RTM simulations and the observations of pyroCb from California Creek Fire on September 5, 2020.

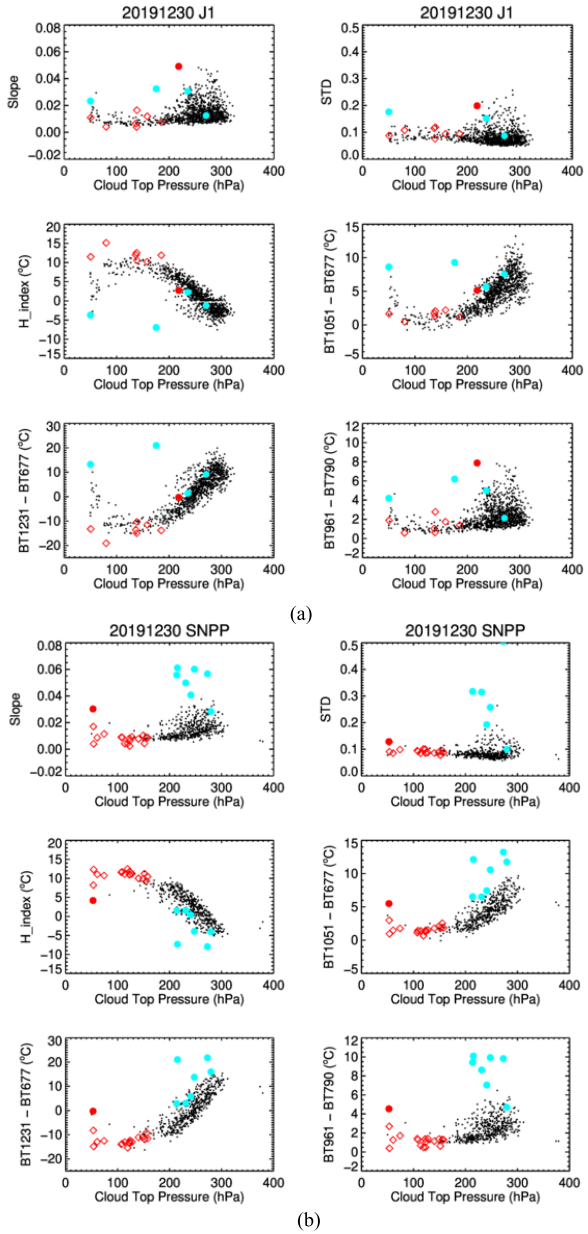


Fig. 15. (a) Variation of characteristic parameters with cloud-top pressure for pyroCb pixels and Cb cloud pixels (black crosses). Red diamonds are for pixels of group (A), red dots are for pixels #7, and cyan dots are for pixels #8–#11. The Cbs are cloud pixels with a retrieved cloud optical depth greater than 3, BT11 below −35 °C, and south of –42°S. (b) Same as (a) but using data from S-NPP. Red diamonds are for pixels of group (A), red dots are for pixels #17 for S-NPP, and cyan dots are for pixels #18–#25.

It was found that for pyroCb pixels with BT11 < −60 °C, the BTs in the spectral region between 760 and 980 cm^{-1} (or 13.3–10.2 μm) are almost constant and the spectra are very smooth. Instead of using a split-window test, we found that applying a linear fitting to the spectra in the region of 800–980 cm^{-1} (or 12.5–10.2 μm) yields slopes and STDs that can be more effectively used for detecting low clouds or clouds with small ice particles; however, they do not exhibit a distinct pattern to differentiate pyroCb from regular Cbs. While the spectra near the ozone absorption band at 9.6 μm display a regular "V" shape under clear skies or over low

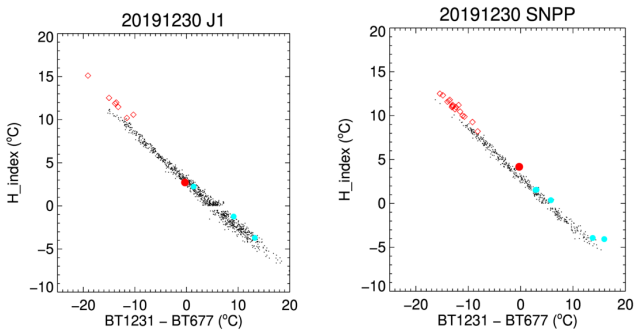


Fig. 16. Scatter plot of BT1231–BT677 versus H_index for the same pyroCb and Cb pixels in Fig. 15. The BT1051–BT677 values for pixel #7 of J-1 and #17 of S-NPP (red dots) are 5.1 and 5.5 K, respectively, suggesting the presence of small ice particles in these two pixels.

water clouds, a pronounced inverted "V" shape is observed for clouds near or above the tropopause. These characteristics were further verified through simulations using PCRTM. A new variable, H_index, was introduced to quantify the height of the inverted "V" near 9.6 μm for cloud-top height detection. A large positive H_index corresponds to clouds near the tropopause, with the largest value occurring at the tropopause, while a negative H_index signifies low water clouds or clear skies. The spectrum begins to invert once the cloud top reaches 200–250 hPa.

Similar to the DCC test using the split-window technique, specific channels from CrIS were used for pyroCb detection. For example, the difference between the maximum temperature near 9.6 μm, i.e., BT1051, and the temperature at a stratospheric sounding channel, BT677, as well as BT1231–BT677, can be used to determine if the cloud top is above or below the tropopause. In addition, BT961–BT790 can indicate the presence of small ice particles in the clouds. The potential application of these tests for pyroCb detection has been explored using pyroCb pixels from 2019 to 2020 Australian fires and further validated with data from California Creek Fires in September 2020. However, the application of these parameters and thresholds on a global scale requires further validation studies.

The single FOV retrieval products, including the retrieved cloud optical depth and cloud-top pressure from SiFSAP, can provide more accurate information about the cloud-top heights and determine whether they are below or above the tropopause. While the mean altitude of these pyroCb pixels is comparable with the findings in [17], we identified five pyroCb pixels with cloud-top heights exceeding 3–6 km above the tropopause. This result indicates that the plume height of this pyroCb, as estimated from an IR sounder, is likely higher than previously reported (e.g., [17]), despite some uncertainties in using standard cloud parameters to represent pyroCb conditions in the retrieval process. Further validation of the SiFSAP cloud-top height against other satellite products, such as Moderate Resolution Imaging Spectroradiometer (MODIS) and VIIRS cloud products, will be important to address these uncertainties and is planned for future work.

Since there are over 20 years of hyperspectral observation data from AIRS, IASI, and CrIS, there is significant potential for a comprehensive historical study of pyroCb by combining MODIS [72] with AIRS, VIIRS with CrIS, and IASI with the Advanced Very High Resolution Radiometer (AVHRR) over the past two decades. The Cloud Profiling Radar (CPR) on CloudSat [73] and the Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP) on the Cloud-Aerosol Lidar and IR Pathfinder Satellite Observation (CALIPSO) [74] also provide cloud-top height observations, and their comparisons with AIRS and MODIS products have been examined in previous studies (e.g., [75], [76]). Using the SiFSAP-retrieved cloud properties, together with the retrieved temperature, water vapor, trace gases (such as CO and ozone) profiles, and surface emissivity, will further help us to gain more insights into the thermodynamic structure of pyroCb. This information will advance our understanding of the formation and lifecycle of pyroCb, facilitating studies on the impact of wildfires and pyroCb on weather, climate, and air quality.

This study highlights the use of the inverted "V" feature near 9.6 μm for detecting pyroCb heights with thermal IR hyperspectral radiance data. However, this inverted spectral feature is also present in other DCCs, thus distinguishing pyroCb from regular Cbs using the information of hyperspectral sensors only still remains challenging in most conditions. However, very high pyroCb can potentially be distinguished from Cbs using a combined approach of BT1231–BT677 and H_index, supplemented by BT1051–BT677. By combining hyperspectral sounders with imagers capable of identifying fire locations, we can improve monitoring of pyroCb events.

Nevertheless, these characteristic parameters derived from hyperspectral IR spectra indeed provide more detailed information than broadband sensors with limited channels, particularly regarding water absorption and the microphysical properties of ice clouds. While the difference in BT between 3.9 μm and the window channel, with the use of broadband satellite sensors [25], can be effectively utilized for pyroCb detection during daytime, this spectral information from hyperspectral IR can add unique value for pyroCb observation during nighttime. It is expected that using hyperspectral IR spectral features along with the SiFSAP retrieval products would greatly enhance the monitoring of pyroCb events and the plume top heights once future hyperspectral sounders and imagers become operational on U.S. Geostationary Extended Observations satellite systems.

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